



UNIMORE
UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

Dipartimento di Ingegneria
“Enzo Ferrari”

Fondamenti di Elettronica

Sensor and Actuators

Laurea Triennale in Ingegneria Informatica

Prof. Francesco Maria Puglisi
email: francescomaria.puglisi@unimore.it
phone: +39-059-2056352

Transducers

Introduction

What is a **transducer**? A device that transforms the variations of a physical quantity in a related variation of an electrical quantity. Translates information from a domain to another.

Typical transducers we will have a look at:

- Temperature
- Force
- Torque
- Position - Velocity - Acceleration
- Magnetic Field
- Current
- Voltage
- Pressure
- Flow
- Level
- Humidity

Transducers or Sensors?

Introduction

- **Sensor**: The primary element in a measurement chain that converts the physical variable information into a signal that is suitable for measurement
- **Transducer**: A device that gets as an input a physical variable and converts it into a variable (of the same or of a different type) according to a well-defined relation

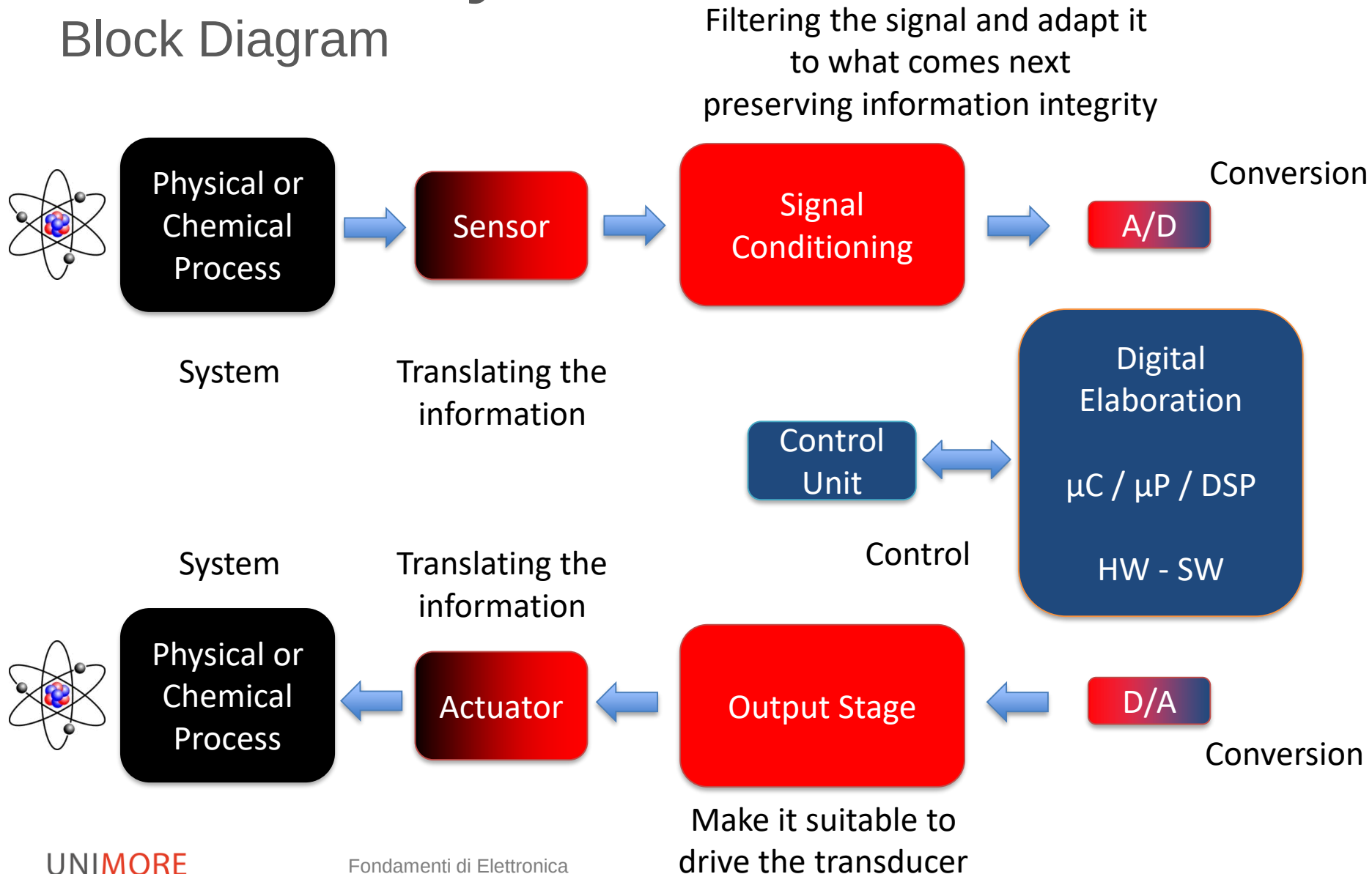
The transducer can be so formed by:

- The sensor itself
- Conditioning circuits
- Elaboration circuits

Nowadays the two terms are used kind of interchangeably

Electronic Systems

Block Diagram



Transducers

Features

Static features of transducers are:

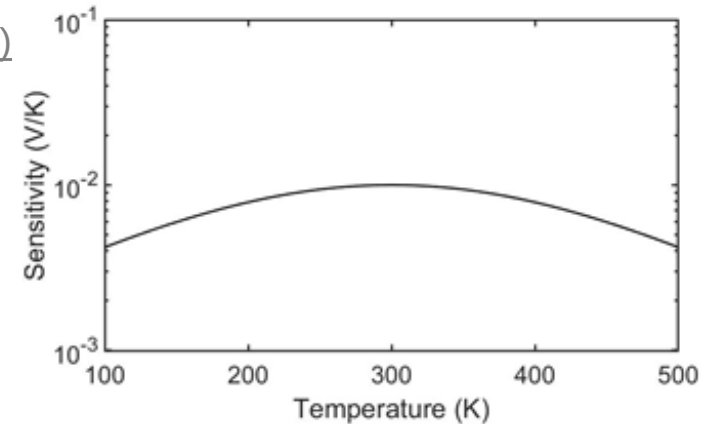
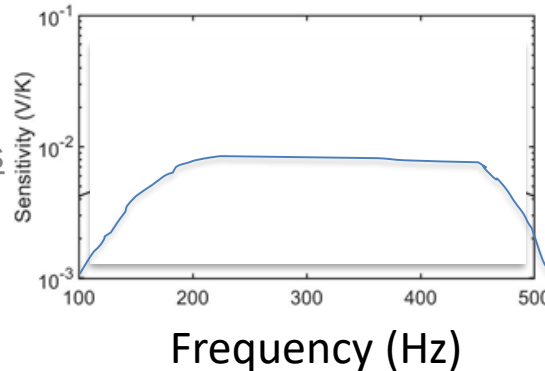
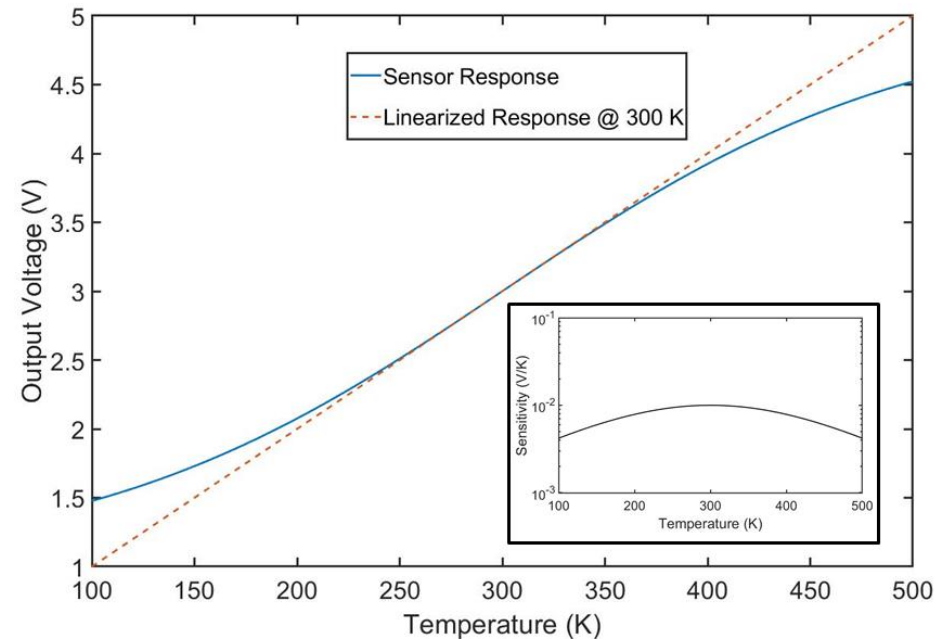
- **Input (span) and output range (full scale)**
- **Sensitivity / gain**
- **Resolution**
- **Linearity**
- Offset
- **Repeatability - Stability**
- Hysteresis
- **Selectivity**
- **Loading effects**

Dynamic features:

- Transfer function
- Step response (time constant, rise/settling time, overshoot)
- Frequency response (bandwidth, damping, natural frequency)

Reliability features:

- Rated lifetime
- Number of full scale cycles
- Fault tolerant



Temperature Sensors

Introduction

We want to measure temperature, so let's think about it

- Temperature flows (heat) by means of conduction, convection, and radiation (this allows contactless)
- Loading effects on the system must be small (low thermal capacity of the sensor)
- The sensed temperature is an average value (over space and over time)
- It is subject to errors

Temperature sensors:

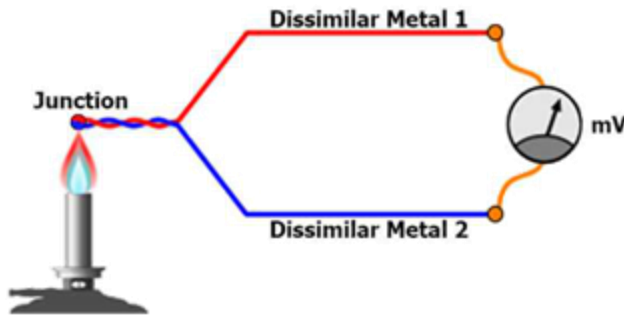
- Thermocouples
- RTD (resistance temperature detectors, a.k.a. thermo-resistance)
- Thermistors (thermal resistors)
- Integrated sensors
- Pyrometers

Thermocouples

Seebeck Effect

They exploit the Seebeck effect

- Conversion of heat directly into electricity at the junction of different types of wire.
- Originally discovered in 1794 by Alessandro Volta, it is named after the Estonian physicist Thomas Johann Seebeck, who in 1821 independently rediscovered it.
- Two different metals joined together, with a temperature difference between the joints will exhibit a voltage difference.
- In the linear approximation, $V = S \nabla T$ where S is the Seebeck coefficient (thermopower).

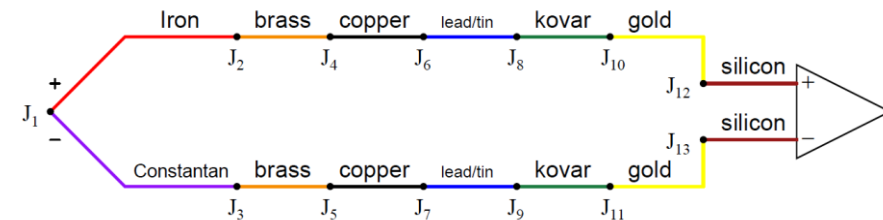
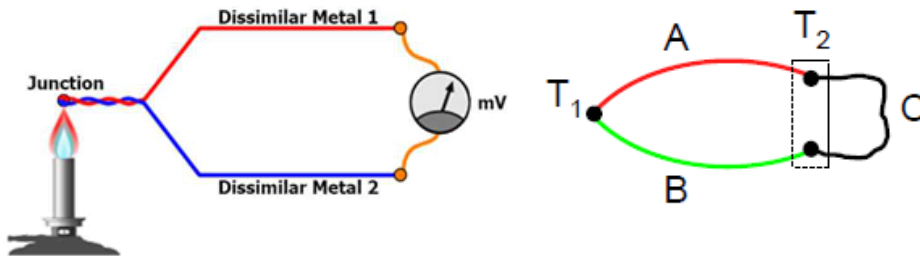


- The Seebeck coefficient generally varies as function of temperature and depend strongly on the composition of the conductor. For ordinary materials at room temperature, the Seebeck coefficient may range between $-100 \mu\text{V/K}$ and $1000 \mu\text{V/K}$
- So thermocouples can measure a temperature difference; an absolute temperature may be found by performing the voltage measurement at a known reference temperature.

Thermocouples

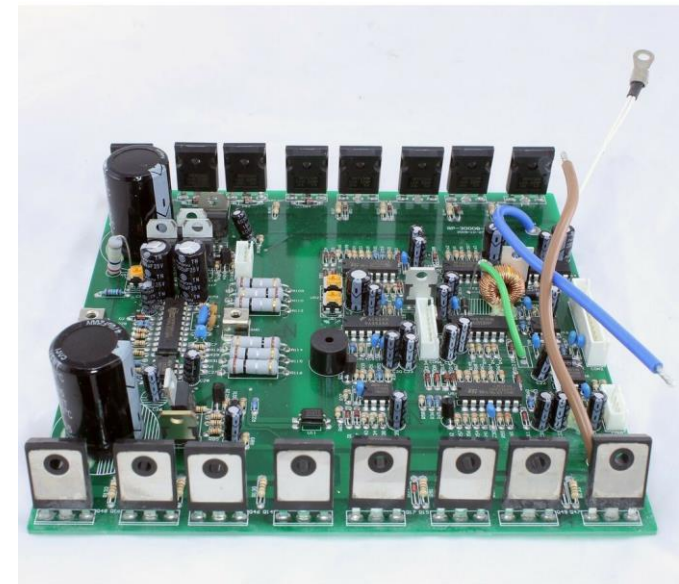
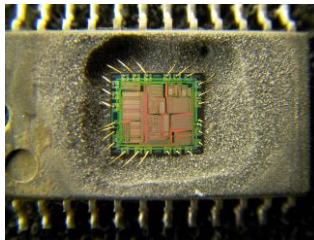
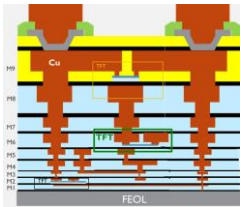
Interposing Metals

Inserting a third metal in the loop does not alter the generated voltage if and only if the new two junctions created by the new metal insertion are at the same temperature. Important for voltage measurements.



Materials are chosen depending on:

- Seebeck coefficient
- Temperature range
- Resistance to harsh environments
- Oxidation rate



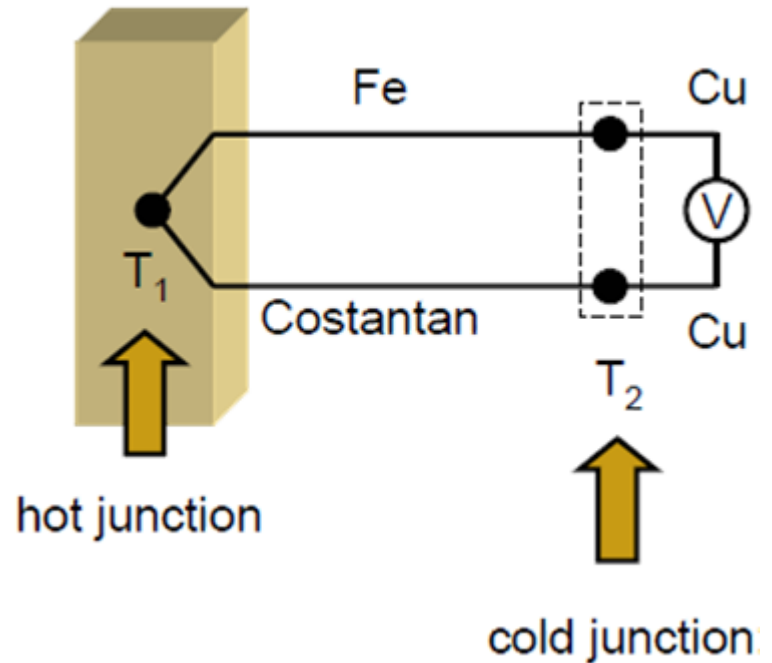
Thermocouples

Cold Junction Compensation

Measuring the absolute temperature at the hot junction requires knowing the cold junction temperature.

Typically two alternatives:

- Ice bath
- Compensation



Constantan is a nickel-copper alloy that consists of 55-60% copper and 40-45% nickel. Its resistivity is constant over a wide range of temperatures. A similar material is manganin ($\text{Cu}_{86}\text{Mn}_{12}\text{Ni}_2$).

Thermocouples

Cold Junction Compensation



LT1025

Micropower Thermocouple
Cold Junction Compensator

Measuring the absolute temperature at the hot junction requires knowing the cold junction temperature.

How do we compensate?

- Typically an IC is used
- Connected to the cold junction
- Features another T sensor
- Can have multiple options to handle many kinds of thermocouples (like E, J, K, R, and so on)

FEATURES

- 80 μ A Supply Current
- 4V to 36V Operation
- 0.5°C Initial Accuracy (A Version)
- Compatible with Standard Thermocouples (E, J, K, R, S, T)
- Auxiliary 10mV/°C Output
- Available in 8-Lead PDIP and SO Packages

APPLICATIONS

- Thermocouple Cold Junction Compensator
- Centigrade Thermometer
- Temperature Compensation Network

DESCRIPTION

The LT[®]1025 is a micropower thermocouple cold junction compensator for use with type E, J, K, R, S, and T thermocouples. It utilizes wafer level and post-package trimming to achieve 0.5°C initial accuracy. Special curvature correction circuitry is used to match the "bow" found in all thermocouples so that accurate cold junction compensation is maintained over a wider temperature range.

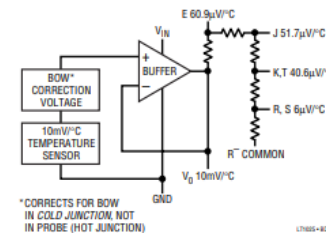
The LT1025 will operate with a supply voltage from 4V to 36V. Typical supply current is 80 μ A, resulting in less than 0.1°C internal temperature rise for supply voltages under 10V. A 10mV/°C output is available at low impedance, in addition to the direct thermocouple voltages of 60.9 μ V/°C (E), 51.7 μ V/°C (J), 40.3 μ V/°C (K, T) and 5.95 μ V/°C (R, S). All outputs are essentially independent of power supply voltage.

A special kit is available (LTK001) which contains an LT1025 and a custom tailored thermocouple amplifier. The amplifier and compensator are matched to allow a much tighter specification of temperature error than would be obtained by adding the compensator and amplifier errors on a worst-case basis. The amplifier from this kit is available separately as LTKA0x.

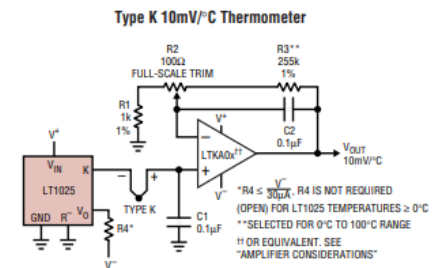
The LT1025 is available in either an 8-pin PDIP or 8-pin SO package for temperatures between 0°C and 70°C.

LT, LTC and LT are registered trademarks of Linear Technology Corporation.

BLOCK DIAGRAM



TYPICAL APPLICATION



Thermocouples

Conclusions

Economic (very inexpensive)

Robust

Can tolerate harsh environments

Stability over time

Wide temperature range

Wide selection of metal pairs

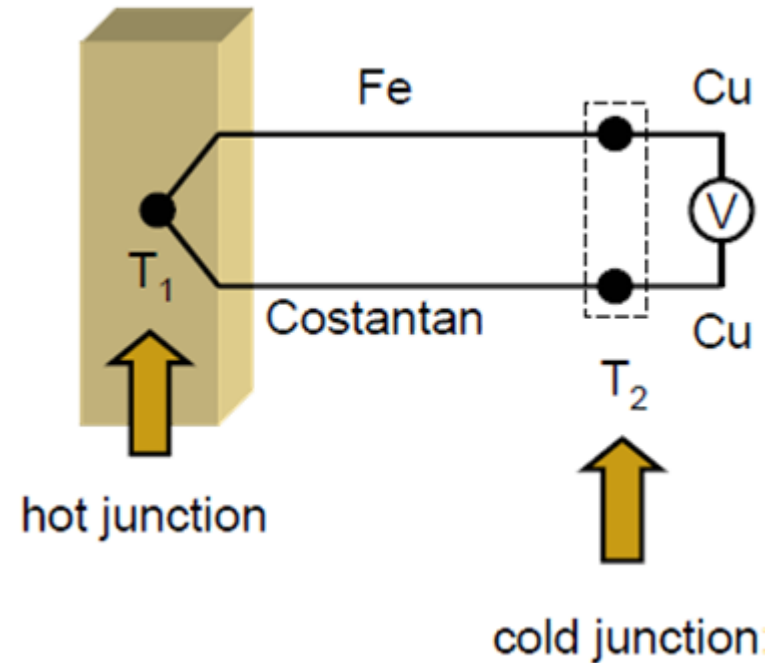
Not so precise (difficult to get systematic errors below 1 K)

Low output voltage (noisy)

Non-linear (polynomial coefficient available in tables)

Typical applications:

- Induction ovens



RTD

Resistance Temperature Detectors

Based on the resistivity temperature dependence in metals

- Pt100, Copper, Nickel
- Very stable
- Precise
- Very Accurate
- Sufficiently Linear
- Expensive
- Suffer from Joule heating
- Slow (thermal inertia)
- 4-wire link required
- Wide temperature range

Pt100

- Accurate and very stable in 0 – 500 °C
- Not attacked by most chemical agents
- High melting point
- Sufficiently Linear
- Very stable over time
- Long-life
- R=100 @ 0°C
- A = 3.54 °C⁻¹ in 250 -350 °C range

$$\rho = \rho_0(1 + \alpha \cdot (T - T_0)) \quad R = \rho \cdot \frac{l}{S}$$

Thermistors

Semiconductors at Play

Based on the semiconductors conductivity temperature dependence

- Precise
- Cheap
- Very non-linear
- Require complex electronics
- Fragile
- Wild resistance variations (high sensitivity)
- Typical applications in:
 - Heating plants
 - Industrial and consumer electronics
 - Telecommunications

Thermistors

PTC and NTC

PTC

- Positive T coefficient
- Silicon-based
- Almost quadratic
- $R \cong R_{amb}(1 + \alpha\Delta T + \beta\Delta T^2)$
- Much more sensitive at high temperatures
- Suffers from self-heating
- Narrow range
- Used for protection from overheating and overloads

NTC

- Negative T coefficient
- Metal-oxides based
- Exponential
- $R \cong \alpha e^{\frac{\beta}{T}} = R_{amb} e^{\beta\left(\frac{1}{T} - \frac{1}{T_{amb}}\right)}$
- Much more sensitive at low temperatures
- Suffers from self-heating
- Very non linear
- High impedance
- Used for low temperature measurements

Integrated Temperature Sensors

Semiconductors at Play, again

They exploit the temperature dependence of the current and voltage in semiconductor devices (diodes, transistors)

- Reliable
- Precise
- Miniaturized
- Linear
- Cheap
- Narrow range
- Used for fridges and temperature compensation in high-end devices
- Can have both current and voltage outputs

Pyrometer

Exploiting Radiation

Infrared sensor that exploits radiation heat spreading

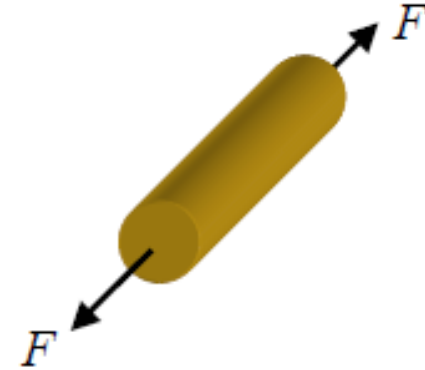
- Proportional to T^4 (very non linear)
- Very stable
- Can withstand high temperatures
- Extended range
- More sensitive to high temperatures
- Expensive

Force Sensors

Introduction

Force sensors:

- Strain gauge
- Piezoelectric



Strain Gauge measures local deformation of a body surface. Upon deformation a metal changes its resistance. In elastic regime we have Hooke's law (uniaxial strain)

T is tension

E is Young's modulus

Piezoresistivity



If non-piezoresistive...

$$T = \frac{F}{A} = E \frac{\Delta l}{l}$$

$$R = \rho \frac{l}{S}$$

$$\frac{dR}{R} = \frac{d\rho}{\rho} + \frac{dl}{l} - \frac{dS}{S}$$

$$\frac{dR}{R} = \frac{dl}{l} - \frac{dS}{S}$$

If properly engineered, ideally...

In reality...

$$\frac{dR}{R} = \frac{dl}{l}$$

$$\frac{dR/R}{dl/l} = 1$$

$$\frac{dR/R}{dl/l} = G$$

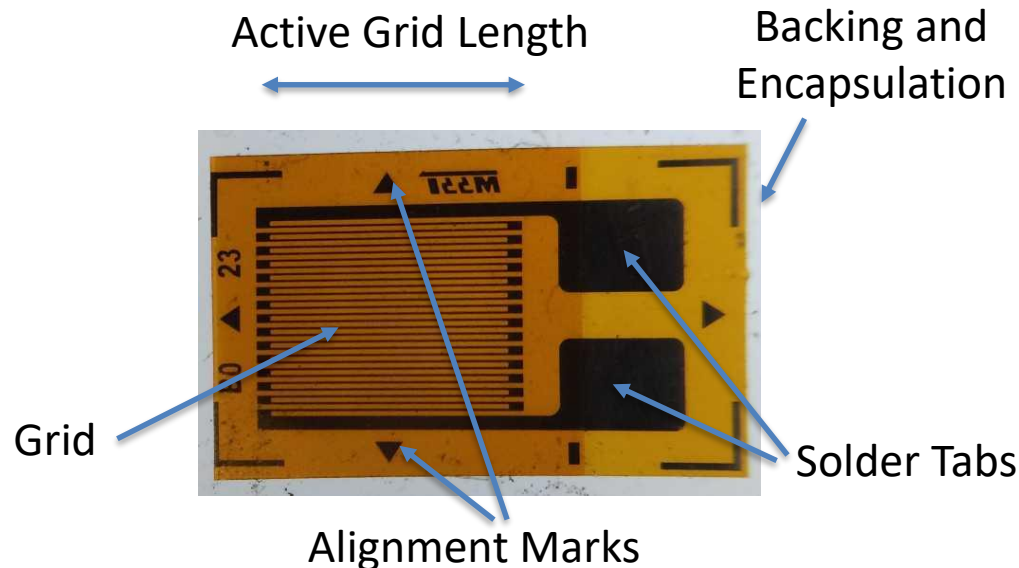
$$\frac{R - R_0}{R_0} = G \frac{l - l_0}{l_0} = G\varepsilon$$

$$R = R_0(1 + G\varepsilon)$$

Force Sensors

Strain Gauge

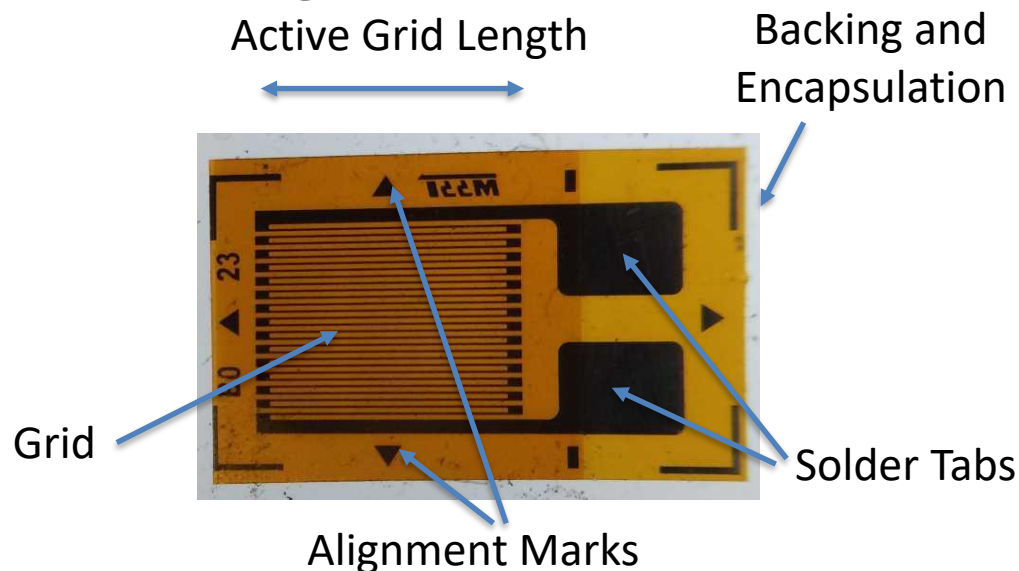
- Strain is generally very small: define the micro-strain ($\mu\epsilon$) = 10^{-6}
- Max deformation in typical strain gauge is 40k $\mu\epsilon$
- For small deformations (first approximation) $R = R_0(1 + G\epsilon)$
- G is the gauge factor (around 2 for most metals, 6 for Pt)
- Mounted (glued) on the surface whose deformation is to be measured



Force Sensors

Strain Gauge

- Resistivity dependence on temperature is critical
 - Use alloys with a low resistance temperature coefficient
 - Use a substrate with the same thermal expansion coefficient
 - Compensation can be done with dummy gauge
- Wheatstone bridge with 2, 3, or 4 wires (wires resistance)
- Op-amp buffered bridge with auto zero cancelation



Force Sensors

Strain Gauge

- Can be used in extreme environments:
 - -269 °C to 1300 °C
 - Radiation Area
 - Vibrations
- Applications:
 - Load Cells, Structures Monitoring (bridges, buildings)
 - Force, Torque, Pressure, Acceleration transducers

Force Sensors

Piezo

- Materials that respond to mechanical stress with a voltage
- Very sensitive
 - Dynamic range in between Hz and kHz
 - Reversible effect (can be used as an actuator)
- With no mechanical stress applied it behaves as a normal dielectric
- Mechanical solicitation deforms the material generating a voltage
- Exploited to realize:
 - Pressure sensors
 - Vibration sensors
 - Microphones, sonars
 - Proximity and level sensors

Torque Sensors

Introduction

Transducer that can measure torque (e.g., torsion of a shaft)

Can be realized exploiting many ideas:

- Strain Gauge
- Inductive
- Magnetic
- Optical

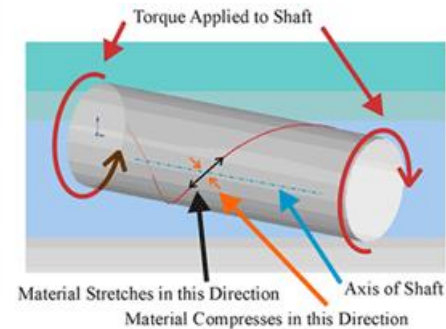
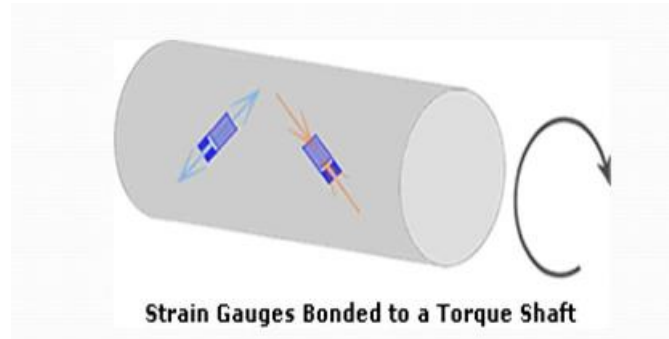
With or without sliding contacts

Static or dynamic



Torque Sensors

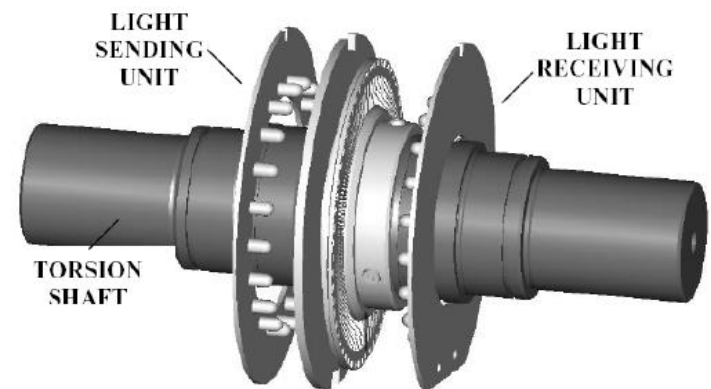
Strain Gauge and Optical Torque Sensors



- Strain Gauge are positioned on the shaft (but it rotates!)
- Optical Torque Sensors exploit light sending and receiving units

Problem: how to supply energy to and retrieve the signal from a rotating mechanism?

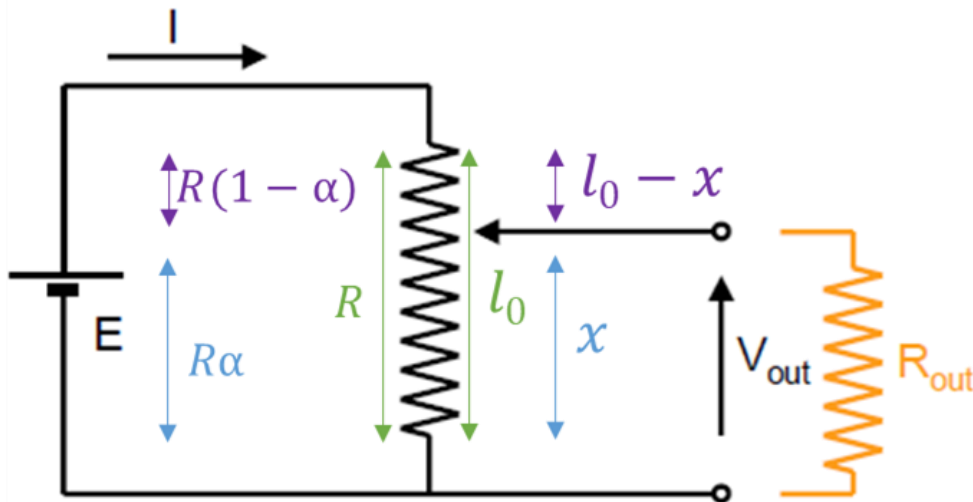
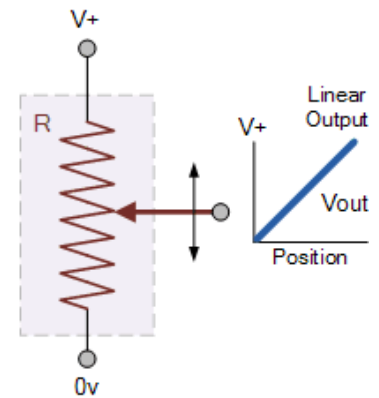
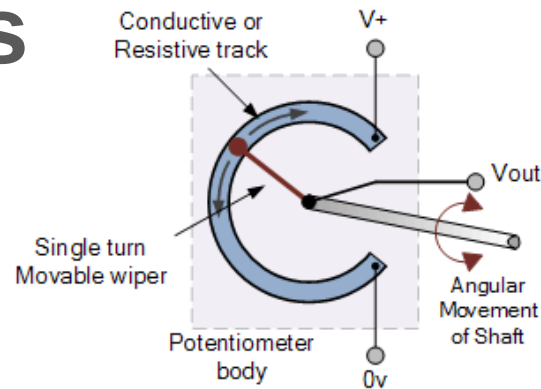
- Wires **X**
- Batteries **V**
- Energy harvesting **V**
- RF (e.g., WiFi) or infrared signaling **V**



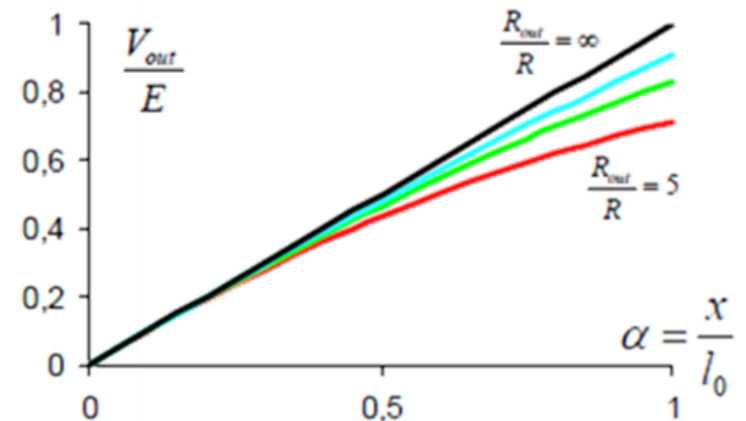
Position Sensors

Potentiometers

- Analog Position Transducer
- Can be linear or rotational
- Can suffer from load effects
- Sliding contacts might be problematic (reliability)
- Very cheap
- Typically not employed in critical position sensing



$$V_{out} = \alpha \frac{R_{out} / R}{-\alpha^2 + \alpha + R_{out} / R} E$$

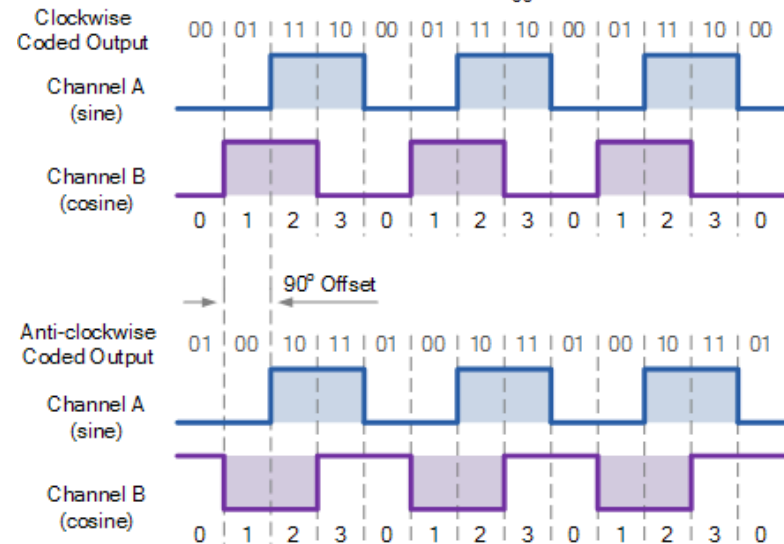
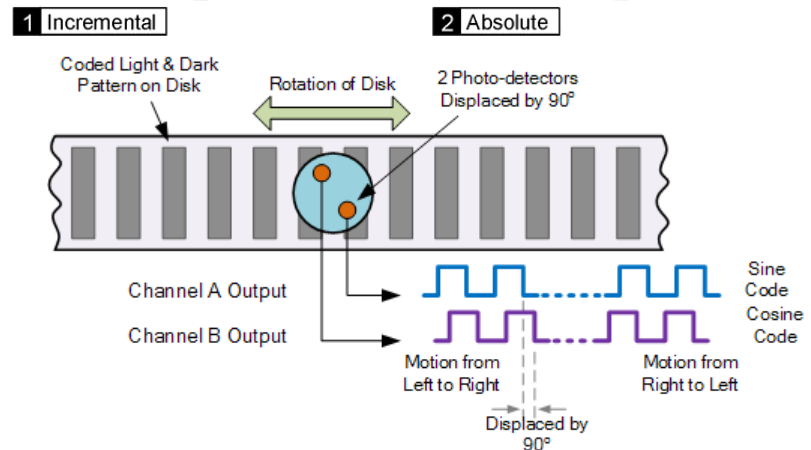
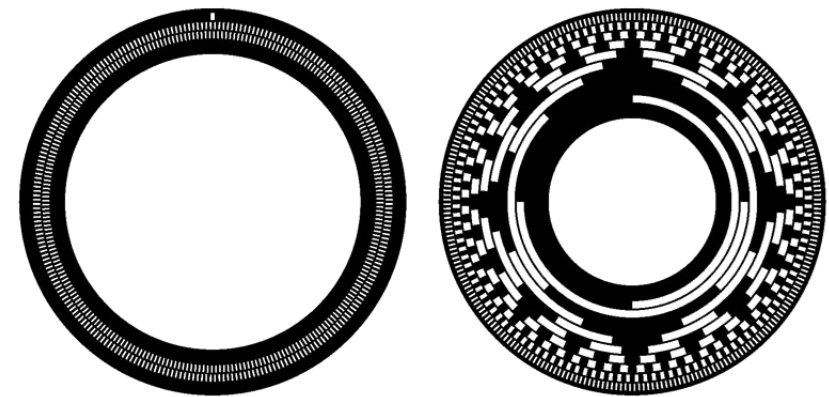
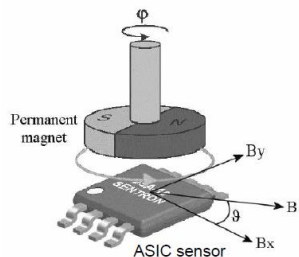


Position Sensors

Encoder and Resolver

Encoder

- Converts angular position in a digital electrical signal
- Linear / rotational
- Absolute or incremental
 - Incremental: needs counter and starter (to set the zero point)
 - Absolute: many codec standards are possible (binary, BCD, Gray, ASCII)
- Typically works up to 80-100 °C
- Mounting requires caution (avoid alignment errors)
- Optical / magnetic
 - Magnetic has larger durability, reliability, operating temperature range. Can have an analog output. High precision. Expensive.



Position Sensors

Encoder and Resolver

Resolver

- Angular position transducer
- Very robust
- Made up of two parts
 - Stator: two windings at 90° from which the output is taken
 - Rotor: rotates as the shaft to be monitored, has 1 winding

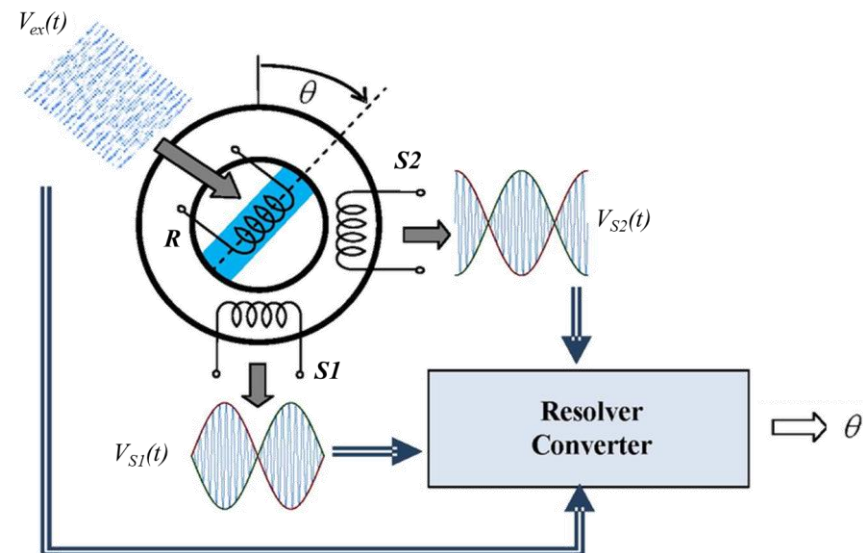
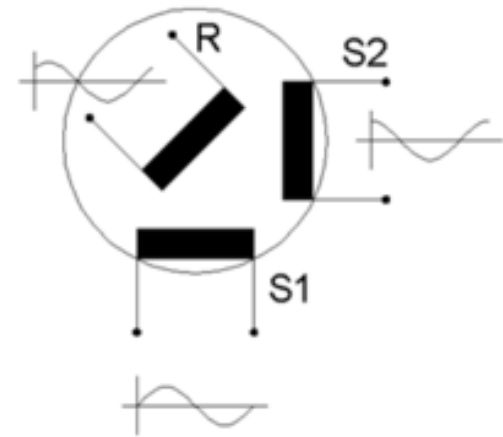
$$V_{gen}(t) = E_0 \sin(\omega t + \varphi)$$

$$V_{ind}(t) = E_0 T \cos(\alpha) \sin(\omega t + \varphi)$$

$$V_{ex}(t) = E_0 \sin(\omega t)$$

$$V_{S2}(t) = E_0 T \cos(\theta) \sin(\omega t)$$

$$V_{S1}(t) = E_0 T \cos\left(\theta - \frac{\pi}{2}\right) \sin(\omega t) = E_0 T \sin(\theta) \sin(\omega t)$$

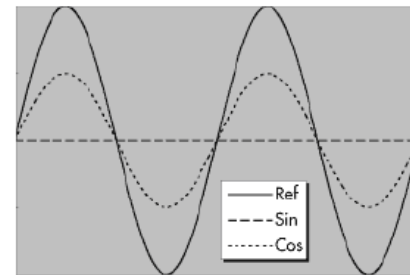
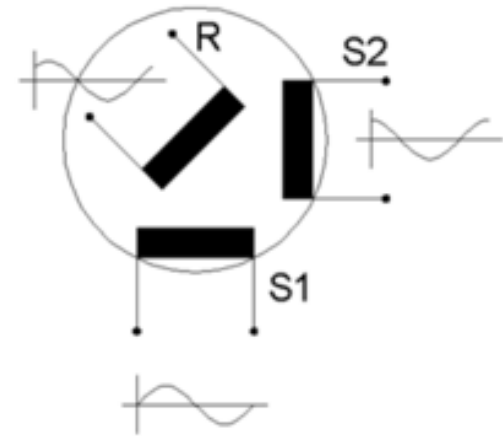


Position Sensors

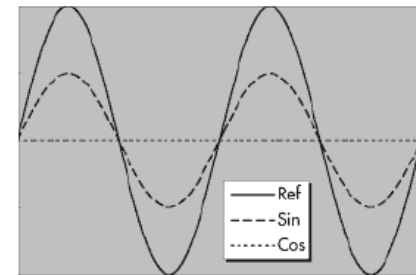
Encoder and Resolver

Resolver

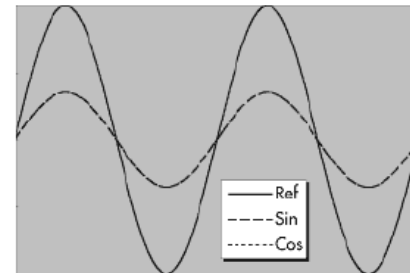
- Angular position transducer
- Very robust
- Made up of two parts
 - Stator: two windings at 90° from which the output is taken
 - Rotor: rotates as the shaft to be monitored, has 1 winding
- Typically biased at 5-15 V_{rms} at 1-15 kHz
- Typically works from -50 up to 150 °C
- Max speed about 10k rpm
- Covers 180° rotation (if no access to the rotor voltage is granted)
 - 360° can be detected with two windings on the rotor or sophisticated read-out circuits



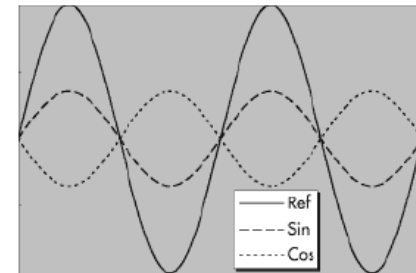
Rotor angle 0°



Rotor angle 90°



Rotor angle 45°



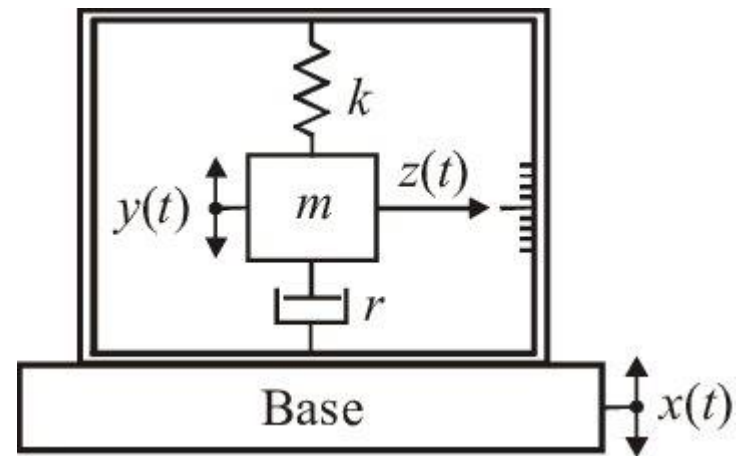
Rotor angle 135°

Acceleration Sensors

Introduction

Use to sense acceleration

- Linear mono-bi-tri axial
- Rotational (MEMS)
- Sensitivity (mV/ms^{-2} or pC/ms^{-2})
- Many kinds:
 - Piezoelectric
 - Piezoresistive
 - Mechanical
 - Capacitive
 - Integrated



Acceleration Sensors

Introduction

Main applications:

- Automotive
 - Airbags
 - Active suspensions
 - Alarms
 - Adaptive brakes
- Crash tests
- Robotics
- Vibration monitoring
- Seismic monitoring
- Impact detectors

Acceleration Sensors

Piezo

Piezoelectric

- Cannot monitor continuous accelerations
- Low damping
- Crystal time constant limits the bandwidth (lower)
- Mechanical resonance limits the bandwidth (upper)
- Special cabling to minimize triboelectric noise
- Large sensitivity
- Large thermal operating range

Piezoresistive

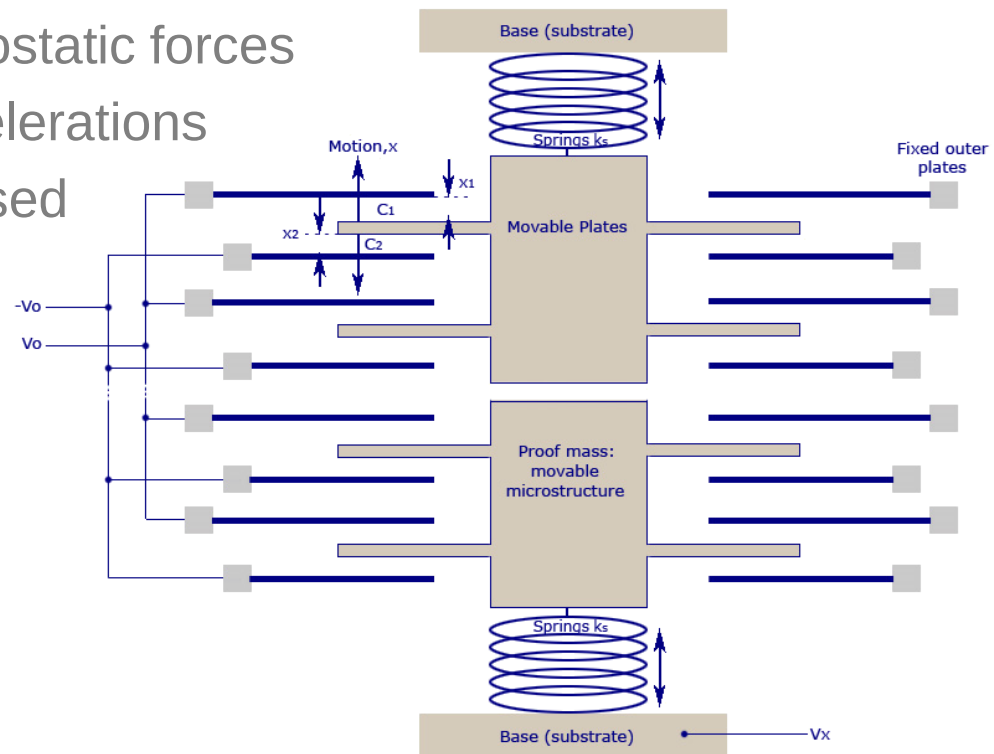
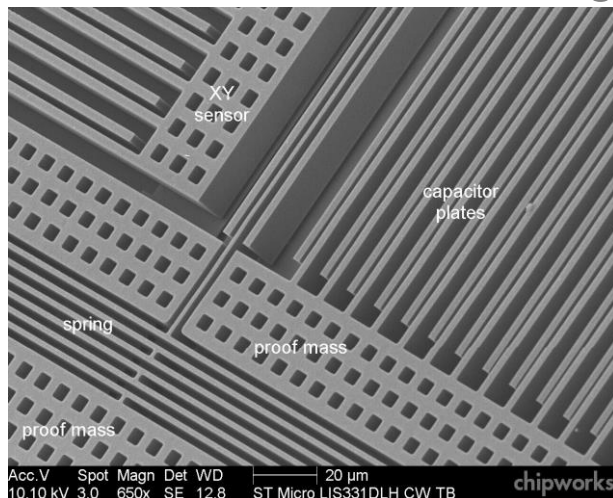
- Strain gauges measure the spring deformation in a seismic system
- Can monitor continuous accelerations
- Sensitive to temperature variations
- Good immunity to electromagnetic fields

Acceleration Sensors

Capacitive

The motion of the mass varies the distance between two plates of a capacitor (tens of microns)

- Capacitance variations are then measured
- Needs to be immune to electrostatic forces
- Can measure continuous accelerations
- Typically two capacitors are used
- Realized in MEMS technology



MEMS Accelerometer

www.InstrumentationToday.com

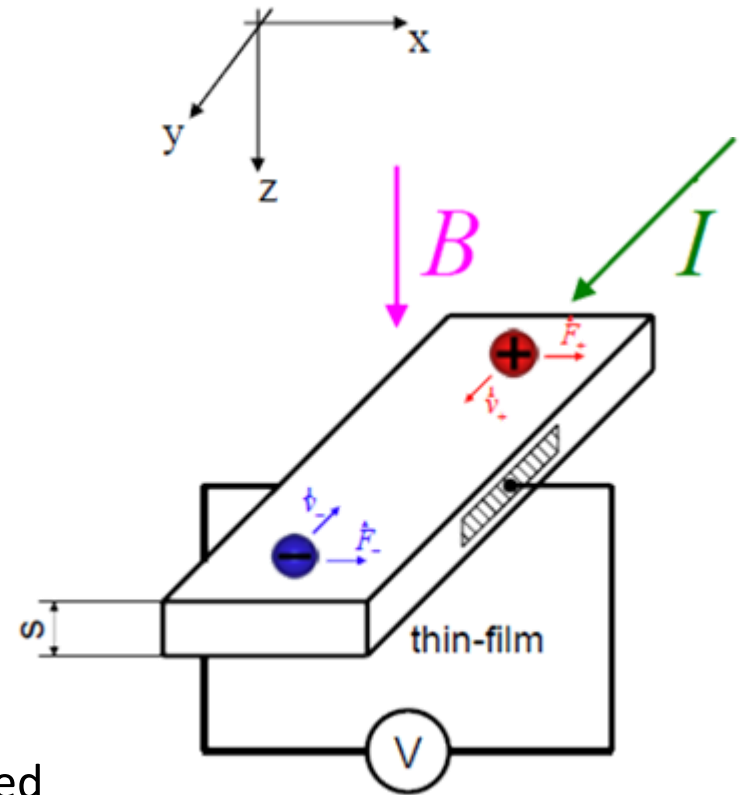
Magnetic Sensors

Introduction

- Hall Sensors
- Magnetoresistive Sensors (MR, AMR, GMR, CMR, EMR)
- Magnetostrictive Sensors
 - The magnetic analogue of the piezoelectric effect

Hall sensors are the most widely employed

- Based on the Hall effect
 - A magnetic field deflects current
 - Induces a voltage that can be measured
- They find application also as current sensors
- Need a magnetic core (permanent magnet or electromagnet)



$$V = KBI = \frac{\mu_q \rho}{s} BI = \frac{1}{nqs} BI$$

$$\frac{V}{I} = R_H$$

Magnetic Sensors

Hall Sensors

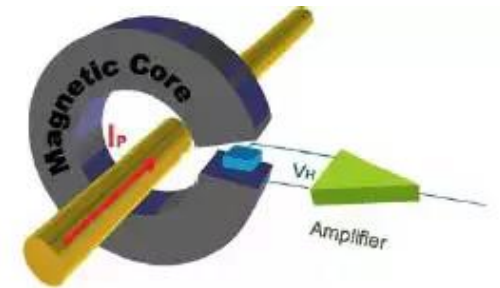
- Bandwidth up to 100 kHz
- Typical application: inverter

Closed-loop current sensor:

Run feedback current in the secondary winding so that the magnetic field generated by the target current is constantly zero.

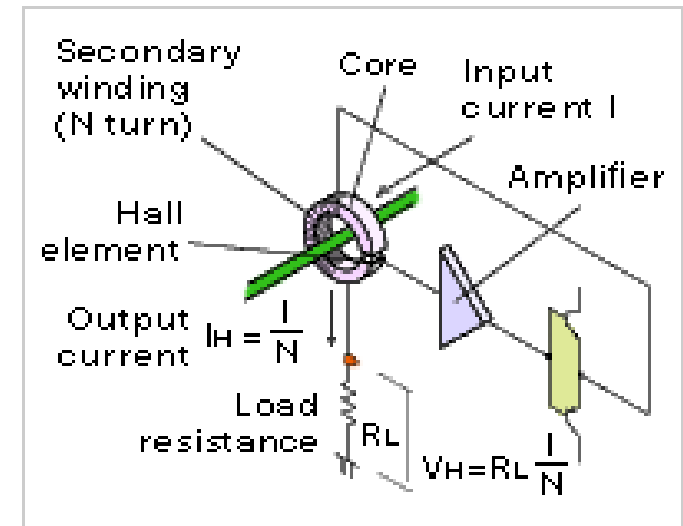
A Hall element is placed in the gap in the core around the input current line, and a secondary coil (N turns) is wound around the core.

The feedback circuit applies a feedback current (I/N) to the secondary coil to cancel out the magnetic field generated by the input current. The load resistor (R_L) converts this feedback current to a voltage, providing a voltage proportional to the input current.



$$V = KBI = \frac{\mu_q \rho}{s} BI = \frac{1}{nqs} BI$$

$$\frac{V}{I} = R_H$$



Magnetic Sensors

Magnetoresistive Sensors

- Some materials can change their resistance as a function of the external magnetic field
- Magnetoresistive sensors are tiny and cheap
- Based always on the Lorentz force
 - Carriers motion is deflected by a magnetic field and as a result the resistance is increased (the actual model says it is the mobility to decrease)

Circa quadratic dependence on B (non-linear)

Large resistance variations

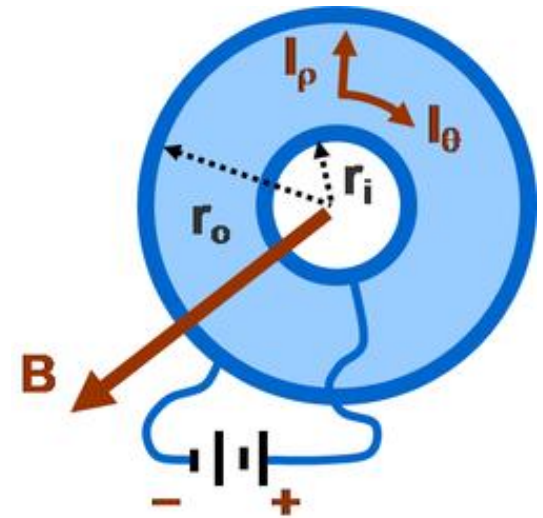
No way to distinguish the B field orientation

Strong temperature dependence (poor selectivity)

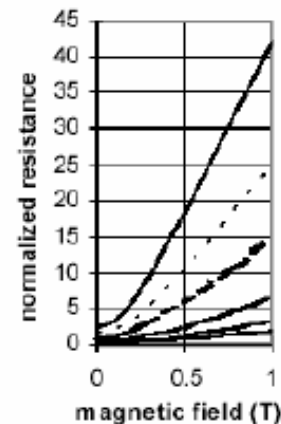
Two sensors can measure B gradients

Can be coupled to permanent biasing magnets

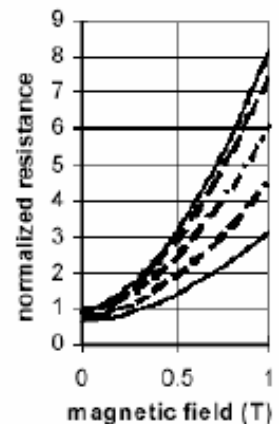
Corbino disc



Siemens D material



Siemens L material



Current Sensors

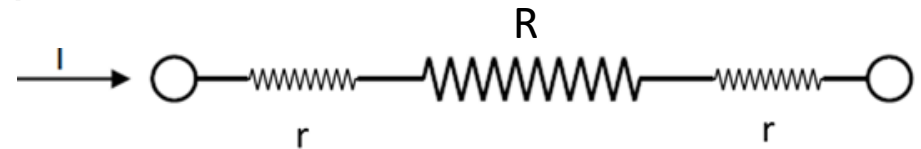
Introduction

- Shunt
- Current Transformer
- Hall probes
- Magnetoresistive probes

Current Sensors

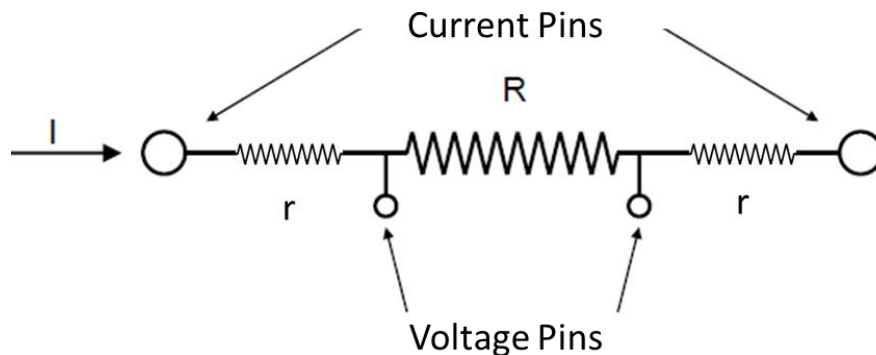
Shunt

- Based on Ohm's law (i.e., $V=RI$)
- Suited for DC and AC (up to high frequencies)
 - Inductance, Skin effect, Proximity effect
- Simple, Robust, Reliable
- Cause a voltage drop (undesired)
- Self-heating problem
- No galvanic isolation



$$\text{calculated } I = \frac{V}{R}$$

$$\text{actual } I = \frac{V}{R + 2r}$$



← 4-terminal resistor

Current Sensors

Shunt

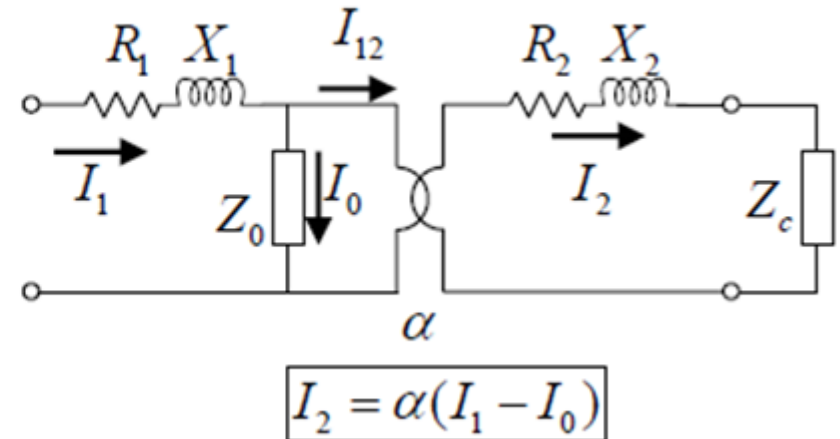
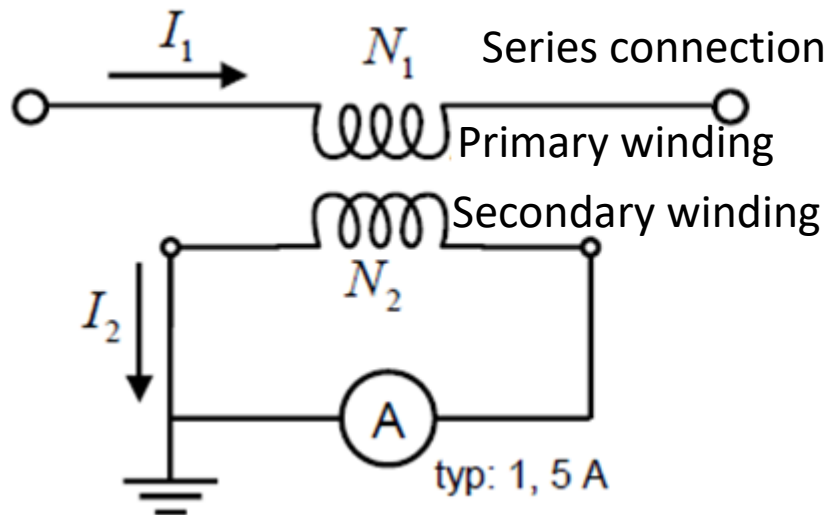


Current rating		Nominal resistance	Power W	I ² t A ² s	Bandwidth MHz	Weight Kg	Type & datasheet
RMS	peak						
8 A	3 kA	200 mΩ*	13	1300	50	1.0	ISM 3P/200
10 A	3 kA	100 mΩ	10	1300	200	1.3	ISM 3P/100
10 A	5 kA	50 mΩ	5	1300	200	1.5	ISM 5P/50
20 A	5 kA	20 mΩ	8	8*10 ³	200	1.5	ISM 5P/20
30 A	5 kA	10 mΩ	9	2*10 ⁴	200	1.4	ISM 5P/10
40 A	5 kA	5 mΩ	8	8*10 ⁴	50	1.4	ISM 5P/5
50 A	20 kA	10 mΩ	25	5*10 ⁴	200	2.5	ISM 50/10
60 A	20 kA	5 mΩ	18	5*10 ⁴	200	2.1	ISM 50/5
70 A	20 kA	2 mΩ	10	5*10 ⁴	200	1.7	ISM 50/2
120 A	40 kA	1 mΩ	14	5*10 ⁴	200	1.5	ISM 100
220 A	40 kA	0.25 mΩ	12	2*10 ⁵	50	1.5	ISM 200
250 A	100 kA	1 mΩ	62	5.6*10 ⁵	200	9.0	ISM 250
360 A	100 kA	0.5 mΩ	65	5.6*10 ⁵	200	9.0	ISM 350
500 A	100 kA	0.25 mΩ	62	2.2*10 ⁶	50	9.0	ISM 500
800 A	100 kA	0.1 mΩ	64	1.4*10 ⁷	8	9.0	ISM 800
1200 A	100 kA	0.05 mΩ	72	9*10 ⁶	12	9.0	ISM 1200
1600 A	100 kA	25 μΩ	64	5.6*10 ⁷	2	9.0	ISM 1600

Current Sensors

Current Transformers

- Transforms a high current in a lower current according to a transformation ratio supposed to be constant
- Used for very high currents (more than 10A)
- Galvanic isolation
- Cannot work in DC



Actuators

Introduction

What is an **actuator**? A device that transforms the variations of an electrical quantity in a related variation of a different physical quantity. Translates information from a domain to another. The aim is to physically act on the system we want to control!

Typical actuators we will have a look at:

- Temperature
- Light
- Force, Displacement, Motion
- Acoustic

Temperature Actuators

Spreading Heat

A **resistor** is a simple thermal actuator!

The device transforms the power dissipated on it as heat, so providing a mean to increase the local temperature.

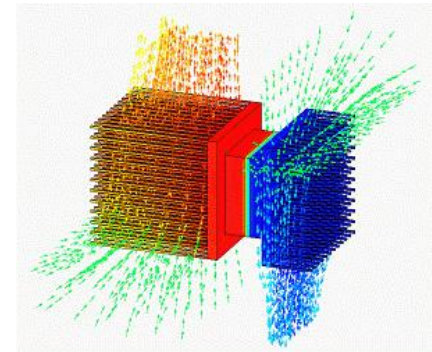


It is a good solution for power levels of few W.

Special structures and heating cables are available to handle much higher power levels.

Important characteristics:

- Thermal Resistance (measured in K/W)
- Rated Power (maximum power that can be dissipated)



What if we want to lower the temperature (w.r.t. ambient T)? Peltier cells!

Let's talk about thermal management (in general) for a while. $T_J = T_A + (R_C + R_J + R_D) \dot{Q}$

Light Actuators

LEDs

A Light Emitting Diode (LED) is a peculiar kind of diode. It is not made of Si (that is an indirect band-gap material), but of direct band-gap materials (like GaAs).



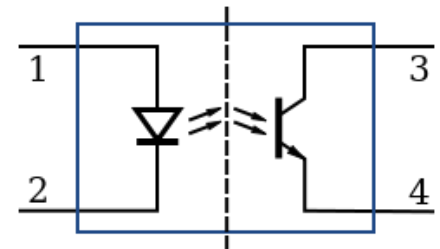
In this way, when this diode is fed with a current, it emits light (and they are very efficient light emitters too).

If properly engineered can emit light in the infrared, visible, ultraviolet.

Typically they have a different working voltage than standard Si diodes.

Important use:

- Lighting
- Digits displays (like the 7-segment)
- Wireless communication (infrared, with phototransistors/photodiodes)
- Opto-isolation (together with phototransistors/photodiodes)



Motion Actuators

Solenoids

A solenoid is composed by an electric coil and a ferromagnetic core that is somewhat free to move within the coil.

Current in the coil will attract the core toward the center of the coil with a force determined by the current in the coil.

Core motion can be constrained by a spring or can be free to move.

Many solenoids are linear actuators, so that the current in the coil generates a motion/force that is linear with the current.

Also available as angular position actuator.



Can be realized with analog-like or digital input

In the digital case, in the ON state the core will move and reach a dead end.

This produces a binary position output, typically used to drive electro-valves, electromechanical devices in which the solenoid uses an electric current to generate a magnetic field to operate a mechanism which regulates the opening of fluid flow in a valve.

Acoustic Actuators

Loudspeakers

An electroacoustic transducer which converts an electrical signal into a corresponding sound. The most widely used is the dynamic speaker.

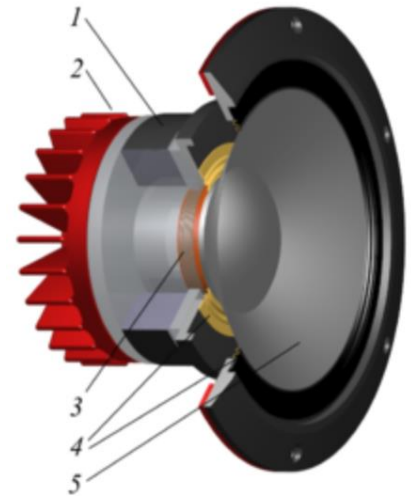
An AC signal is applied to a coil suspended in a gap between the poles of a permanent magnet.

The coil is forced to move rapidly back and forth due to Faraday's law, which causes a diaphragm (usually conically shaped) attached to the coil to move back and forth, pushing on the air to create sound waves.

The sound source (e.g., a microphone) must be amplified with an audio power amplifier before the signal is sent to the speaker.

At higher frequencies (e.g., ultrasound) the magnet is typically replaced by a piezoelectric actuator.

1. Magnet
2. Cooler
3. Coil
4. Suspension
5. Diaphragm

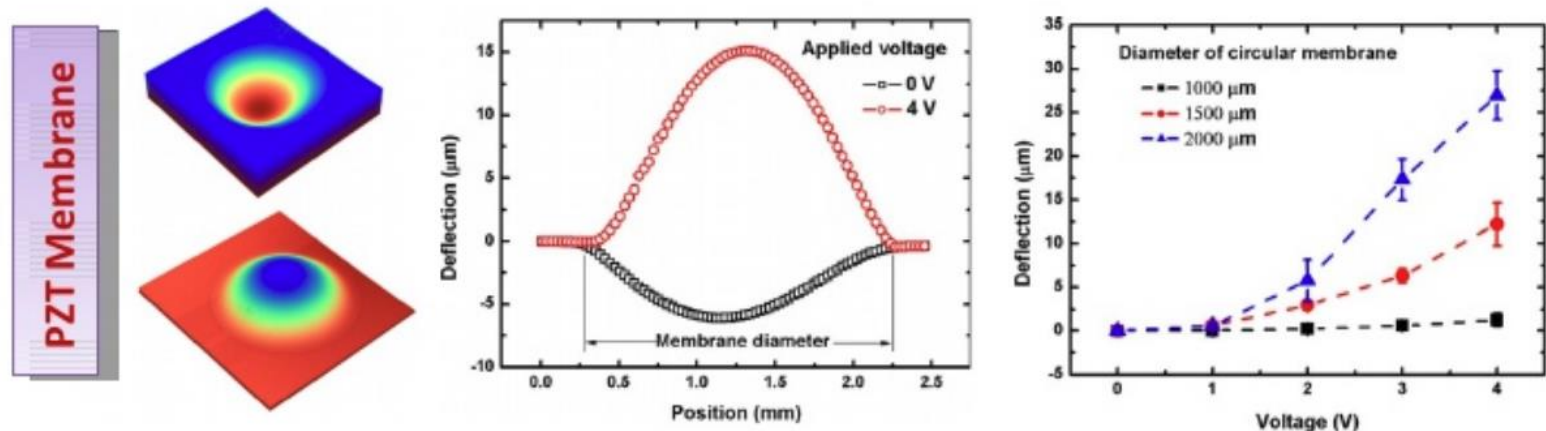


Piezo Actuators

MEMS

A piezoelectric speaker (also known as "piezo", buzzer, crystal/beep speaker) is a loudspeaker that uses the piezoelectric effect for generating sound. The initial mechanical motion is created by applying a voltage to a piezoelectric material, and this motion is typically converted into audible sound using diaphragms and resonators.

Typically they operate well in the range of 1-5kHz and up to 100kHz in ultrasound applications (imaging, obstacle detection, assisted parking, ADAS).



H Bridge

Controlling Motors

$$V = L \frac{dI}{dt}$$

A DC motor can be controlled by a Full Bridge (also called H bridge)
It allows control over the motor direction (right or left), and allows free run (coasting) or braking

The switches of the same leg cannot be closed simultaneously to avoid short circuits

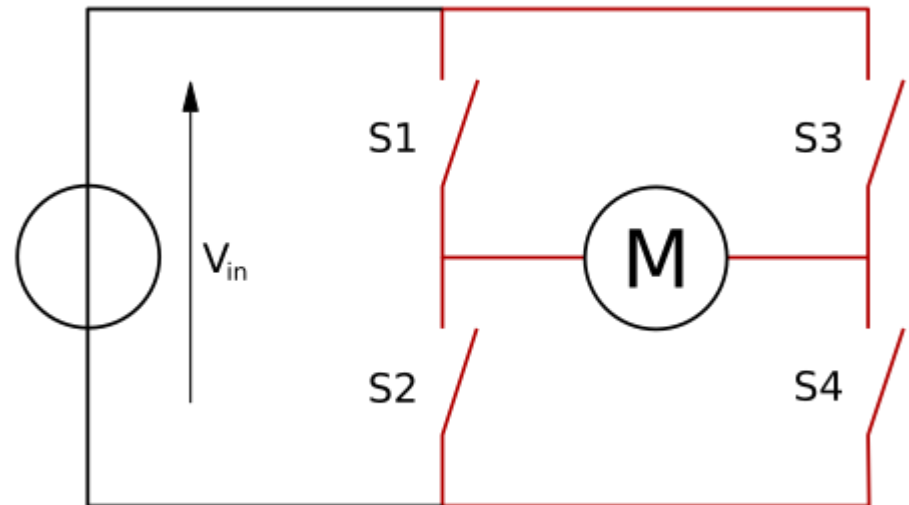
Typical configurations:

S1 and S4: motor moves R

S2 and S3: motor moves L

S2 and S4: motor brakes

Open switches: motor coasts



H Bridge

Controlling Motors

Switches are typically implemented as MOSFETs with a freewheeling diode (typically DC motors are seen as an inductive load)

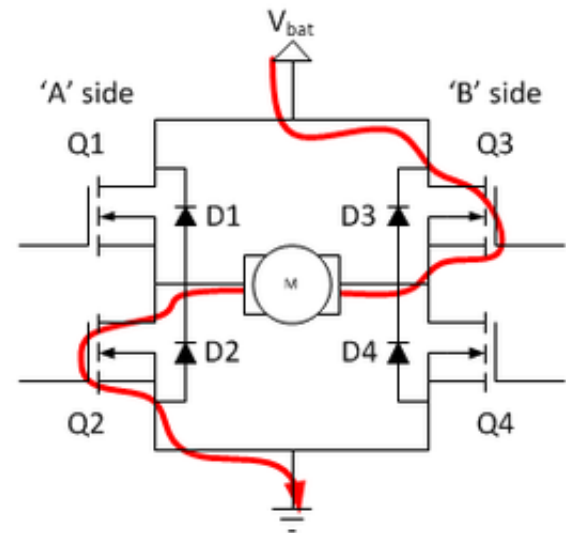
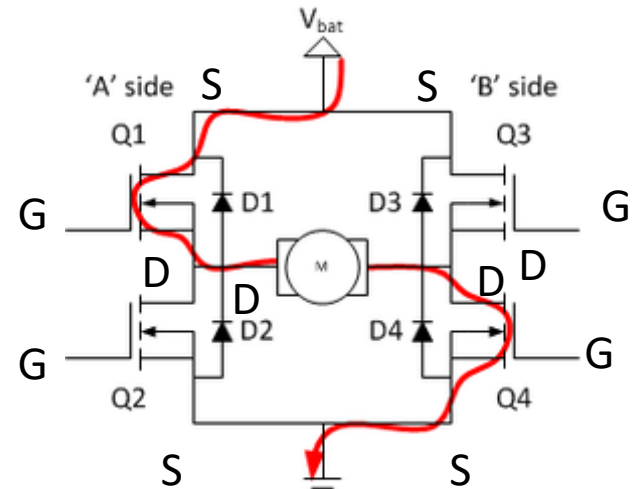
nMOSFETs are used for low-side switching

pMOSFETs are used for high-side switching

nMOSFETs are faster than pMOSFETs!

Can be used also for high-side switching provided that gate drivers with charge pump are used

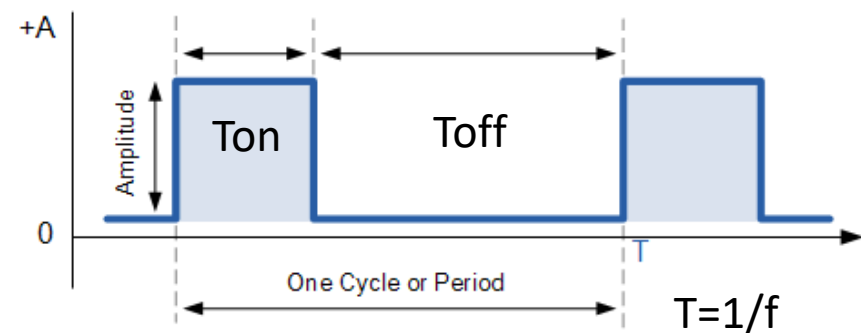
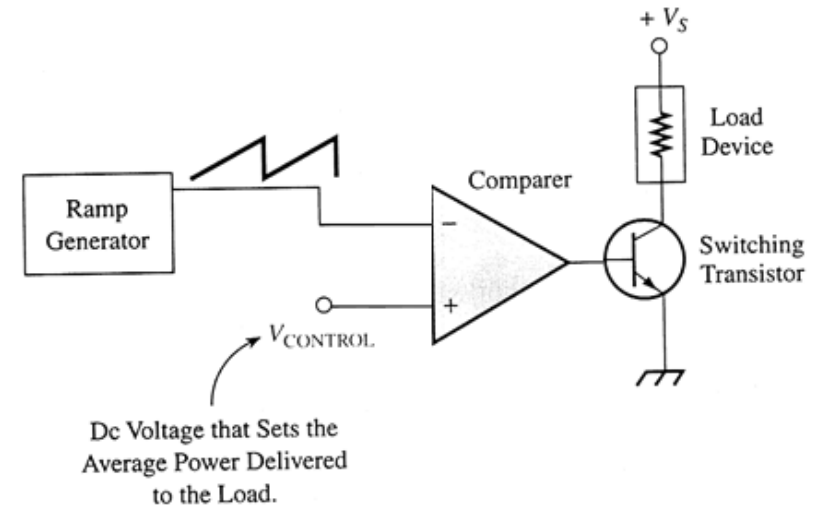
(since the gate of high-side FETs is not grounded)



Pulse Width Modulation

Circuits to handle power

- Typically used to control small electric engines
- A three part system:
 - Ramp voltage Generator
 - Comparator
 - Switching power device
- OFF: no voltage on the load
- ON: all power supply to the load
- The average voltage to the load is the product of the duty-cycle and V_s
- Controlling the duty-cycle we can control the average voltage (and power) delivered to the load
- Also, **employed to generate arbitrary waveforms** using a low-pass filter and a μC that generates an appropriate control signal



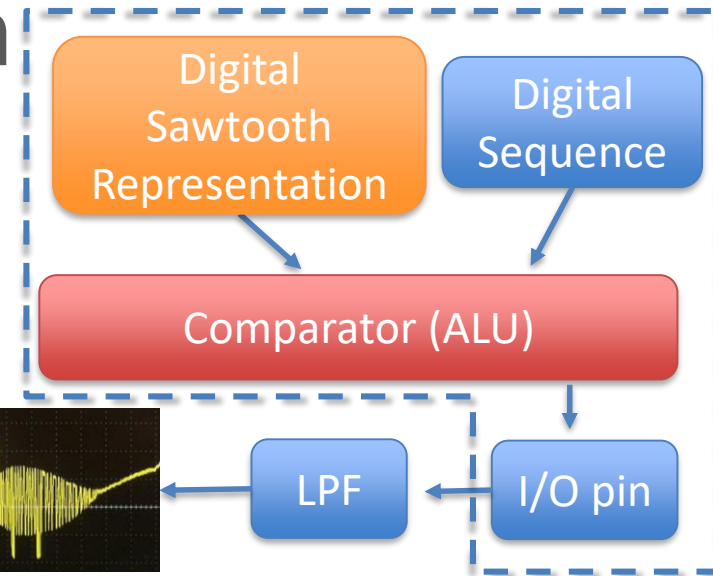
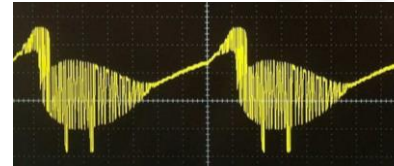
$$D = \frac{T_{on}}{T} \quad P_{av} = D V_s \frac{V_s}{R} = D P_{max}$$

Pulse Width Modulation

Generating and modulating signals

- To generate arbitrary waveforms from digital data:

- The control signal can be generated by a digital sequence
- The ramp signal is also needed
- The two signals (control and ramp) are subtracted with the ALU
- The result is a PWM signal (square wave the duty cycle of which is proportional to the control signal actual value) that controls an I/O pin
- A low-pass filter lets only the average go out



- As a signal modulator:

- The control signal is the input analog signal
- The sawtooth is the «carrier»
- The result is a digital and modulated signals whose duty cycle is proportional to the control signal

